

Measurement Fundamentals

Week 5 — Tuesday Lecture

Today

- Accuracy, precision, and resolution — three different things
- Where measurement error actually comes from
- How much accuracy a decision needs
- Reporting a number you can defend
- Checking it against something you trust

The question this course asks

"The lot is 210 feet long" — an assertion.

"210 ± 6 feet" — a measurement.

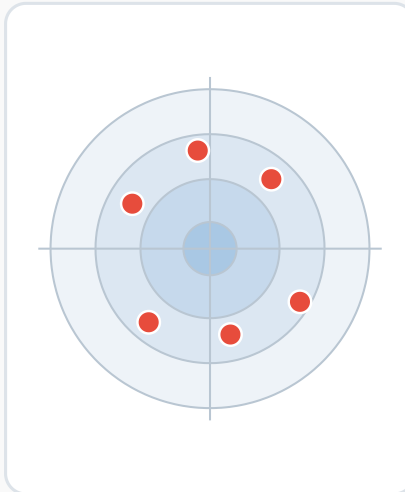
Every method in this course, from pacing to a drone, is judged by the same test.

How good is this number?

You will not know until you can answer that with a range, not a single value.

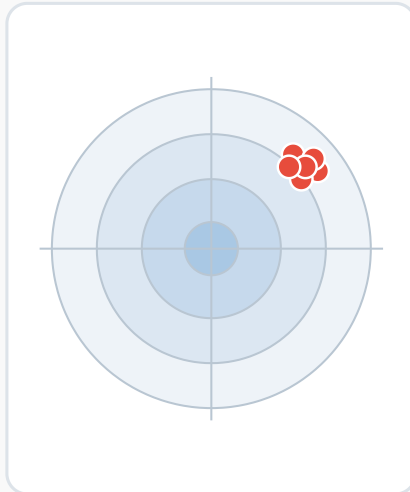
Accuracy and precision are two different things

The bullseye is the true value. Each dot is one measurement of the same thing.



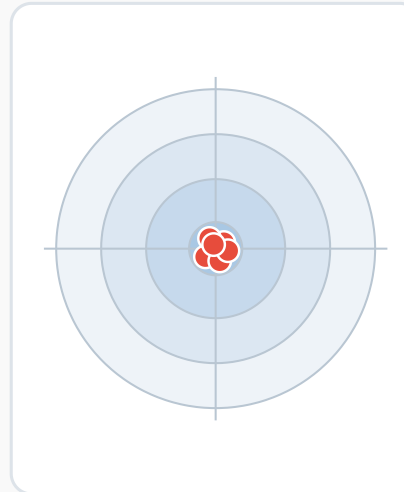
Accurate, not precise

Right on average, badly scattered



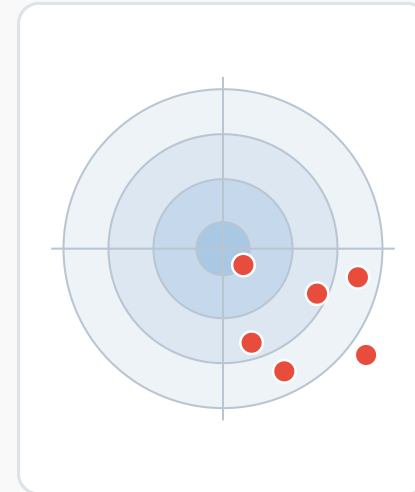
Precise, not accurate

Repeatable, and repeatably wrong



Accurate and precise

What you are aiming for



Neither

Scattered, and off as well

Averaging fixes scatter. It does not fix an error that pushes every measurement the same way.

That is why you need both numbers, and why one of them cannot stand in for the other.

Accurate means close to the true value. Precise means the repeats agree.

Three different things

- **Accuracy** — how close to the true value. Errors are usually **systematic**.
- **Precision** — how closely repeats agree with each other. Errors are usually **random**.
- **Resolution** — the smallest difference the instrument can show you.

Averaging fixes scatter. It does not fix a result that is consistently off.

Precise, and wrong

A 100-ft steel tape has stretched half a percent. Laid along a true 100.00 ft it reads:

99.50 99.52 99.49 ft

Property	This tape
Resolution	1/16 in — excellent
Precision	$\sigma \approx 0.2$ in — excellent
Accuracy	every reading 6 in short

Nothing in the three readings tells you. Only a different tape would.

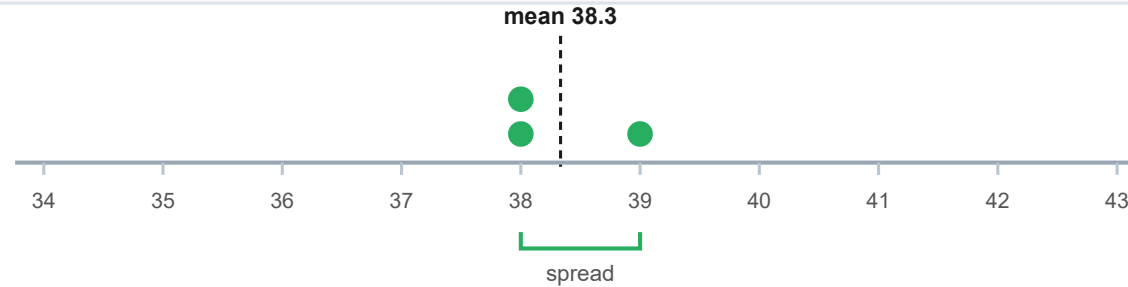
Which walker would you trust for a single walk?

Walker A: 38, 39, 38 steps Walker B: 35, 42, 39 steps

Precision is the spread, and it has a number

Two students pace the same 100 ft three times. Their step counts:

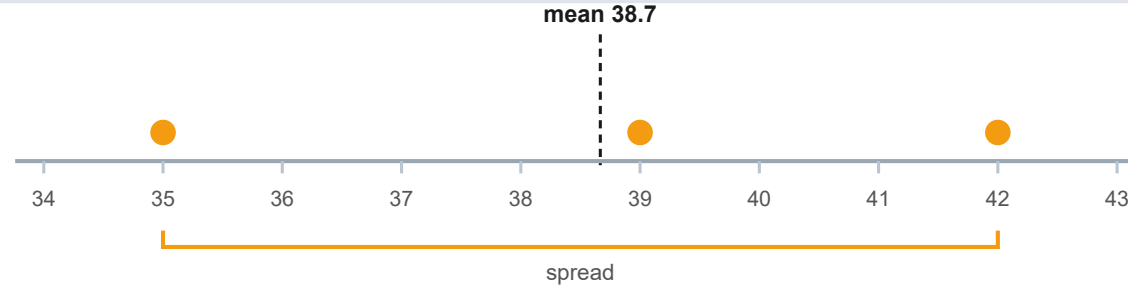
Walker A



$\sigma = 0.6$ steps

precise

Walker B



$\sigma = 3.5$ steps

not precise

Both walkers average about the same count. Only one of them can tell you how far to trust a single walk.

This figure is only about precision. Accuracy needs the tape.

Same average, very different spread — only one walker is precise.

Walker A's standard deviation

$$\bar{x} = \frac{38 + 39 + 38}{3} = 38.33$$

$$\sigma = \sqrt{\frac{(-0.33)^2 + (0.67)^2 + (-0.33)^2}{3 - 1}} = \sqrt{0.33} \approx 0.6 \text{ steps}$$

A tight σ on an average near 38 — Walker A is precise. This says nothing yet about accuracy.

Four places the error comes from

Only one of them is the instrument's fault.

The instrument

A tape that has stretched, or a scale too coarse

→ Check it against a known length

The method

Treating a lot that is not rectangular as a rectangle

→ Match the method to the shape

The operator

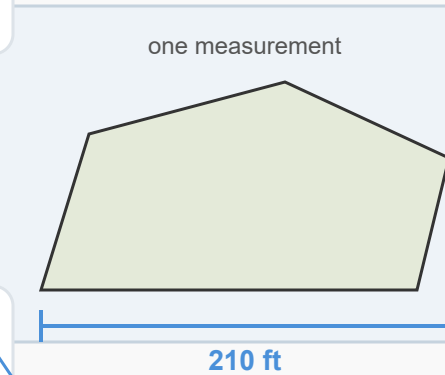
Stretching your stride while calibrating a pace

→ Repeat the measurement, then average

The reference

A benchmark that moved, or imagery three years old

→ Use a reference you can verify



Repeating a measurement exposes the operator. It hides the other three.

Instrument, method, operator, reference — repeating a measurement only exposes one.

Trace it three times

The same lot in Google Maps, three careful traces:

27,600 28,100 27,850 ft² $\sigma \approx 250 \text{ ft}^2$, about 1 %

That spread is the **operator** — how steadily you follow the edge.

The date in the corner of the image is the **reference**. If the lot was restriped after the photo, every trace is wrong the same way.

Which of the four boxes does careful tracing fix? Which does it not touch?

Turning a percent into a range

A calibrated pace is good to about **3%**.

$$3\% \times 210 \text{ ft} \approx 6 \text{ ft}$$

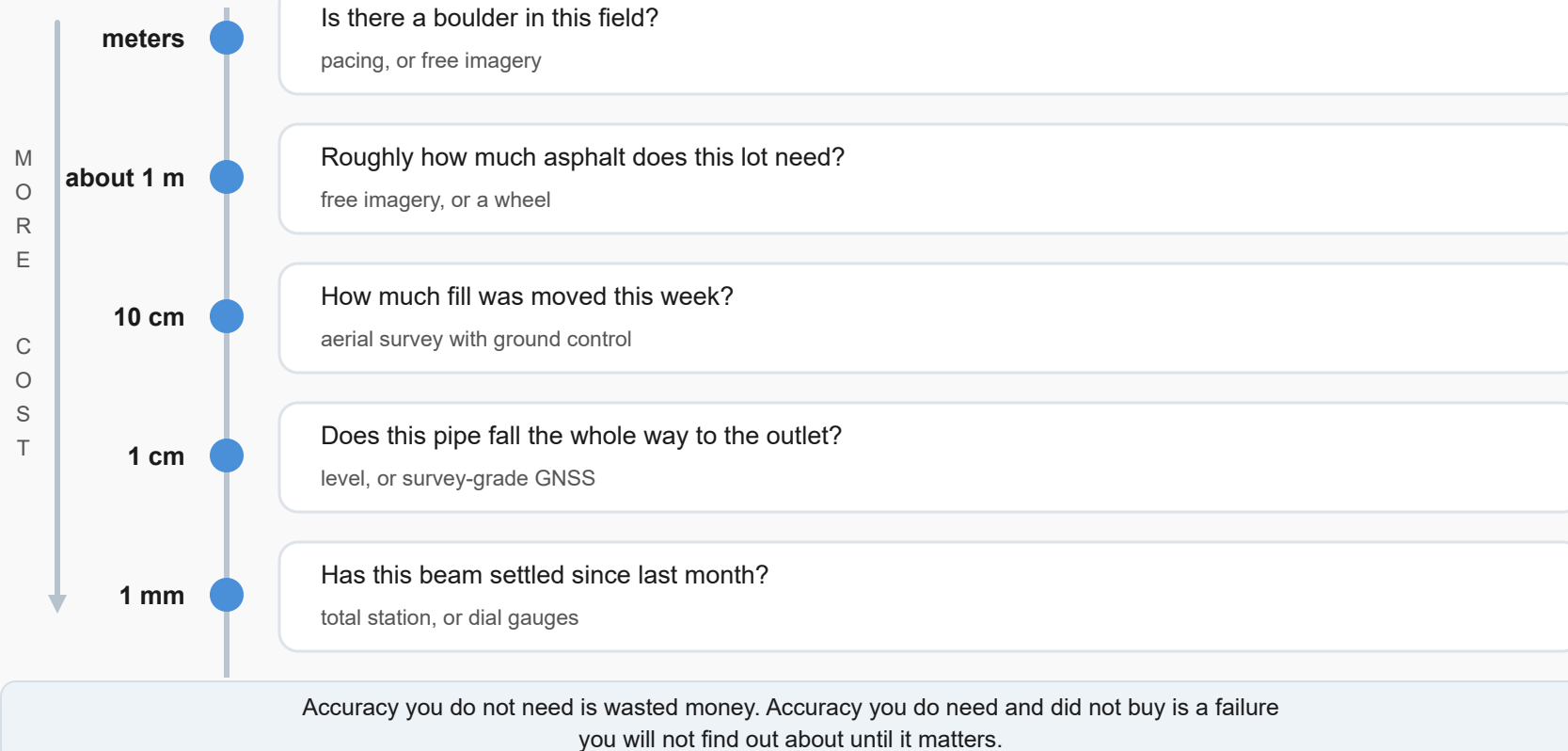
Report **210 ± 6 ft** — not a bare 210.

Boulder, or beam?

"Is there a boulder in this field?" and "has this beam settled?" do not need the same accuracy.

The decision sets the accuracy, not the equipment

Work down the list only as far as the question actually requires.



The decision sets the accuracy needed — meters down to millimeters.

1.2 inches over two acres

A pad is flown before and after a week of grading. Every elevation in the second flight is **1.2 in (3 cm) high** — invisible at any one point.

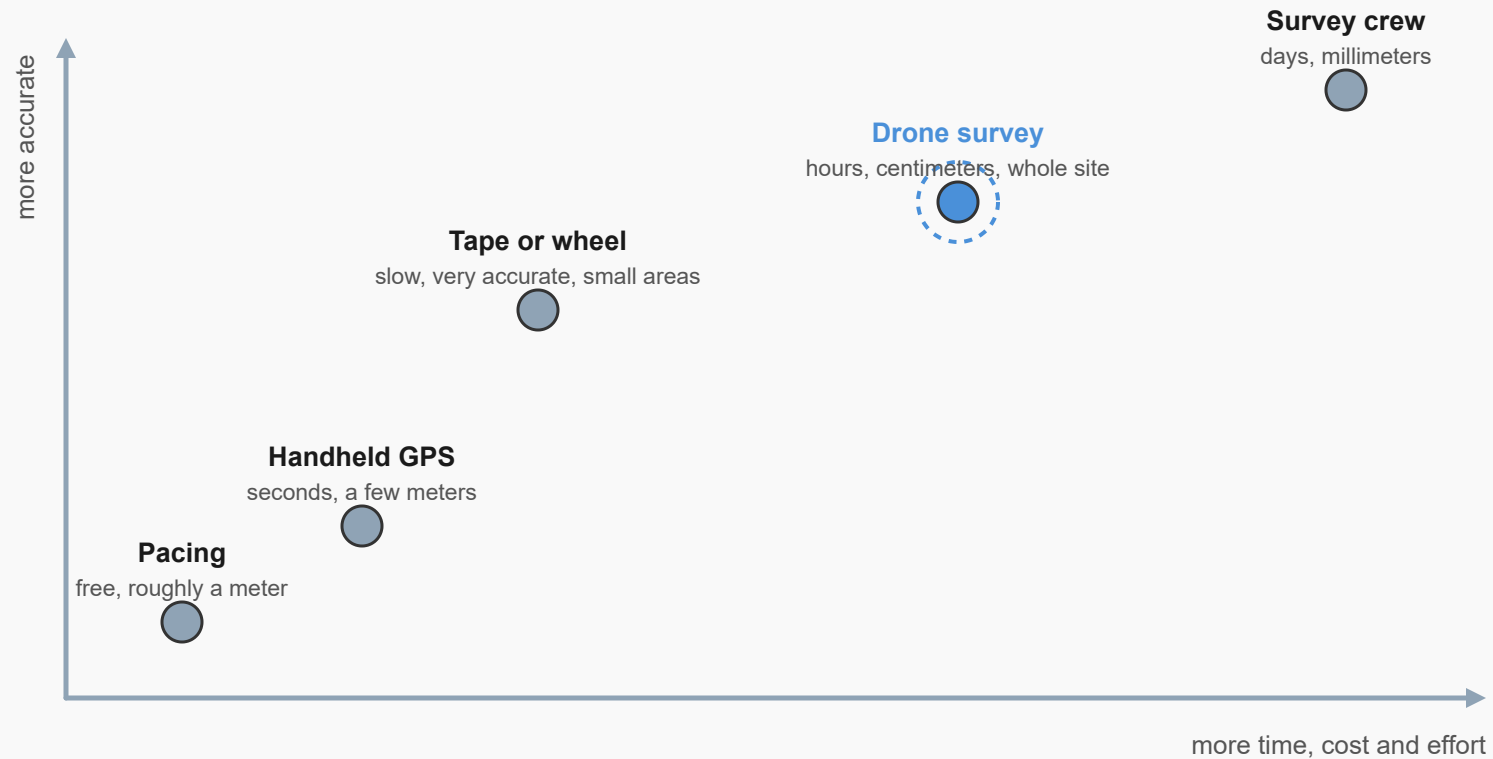
$$87,000 \text{ ft}^2 \times 0.1 \text{ ft} \approx 8,700 \text{ ft}^3 \approx 320 \text{ yd}^3$$

About **thirty truckloads** of fill that was never placed, at the contractor's unit rate.

A small systematic error, multiplied by a large area.

Picking a measurement method

There is no best method, only the cheapest one that still answers the question.



This course is about knowing which dot to reach for, and being able to explain why.

Accuracy rises with effort, but not smoothly — pick the method the decision needs.

Pace it off, or fly it?

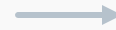
A pair of boots and a drone both answer some question correctly, and both answer some question wastefully.

Your calculator does not know how good your measurement is

42 paces at 2.53 ft per pace, with a pace good to about three percent.

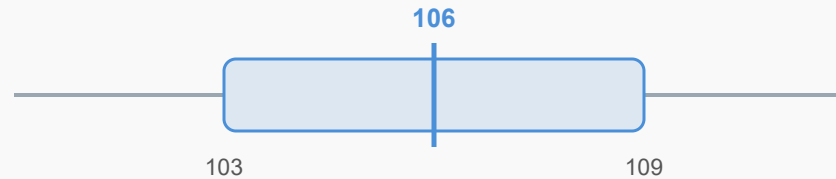
what the calculator says

106.26
decoration



what you report

106 ± 3 ft



three percent of 106 ft is about 3 ft, so the answer lives anywhere in here

Quote no more digits than your uncertainty supports, and put the uncertainty next to the number.

Extra digits do not make a measurement look careful. They make it look unexamined.

Your calculator does not know how good your input was.

106.26, or 106 ± 3 ?

42 paces at a calibrated 2.53 ft/pace:

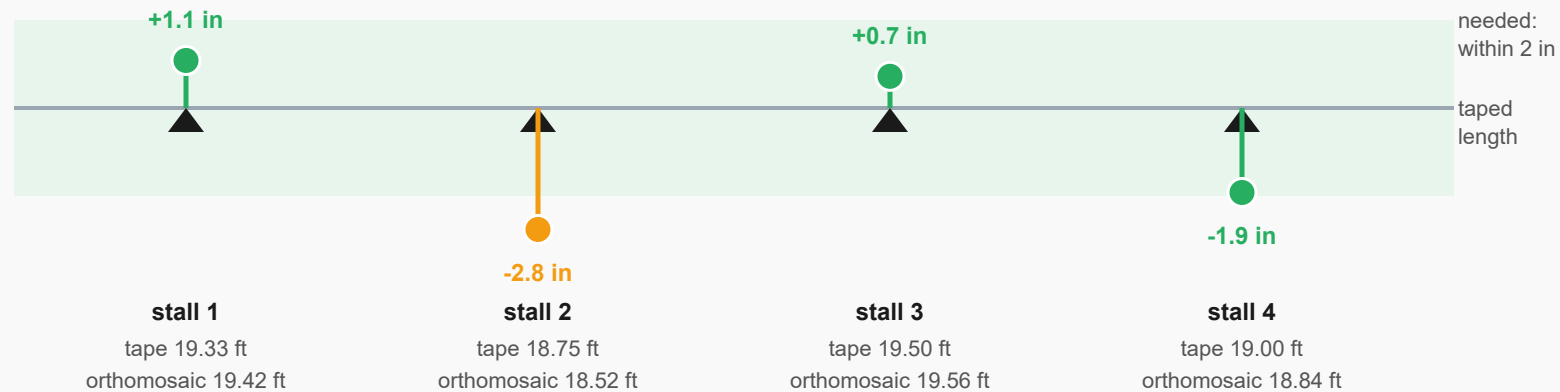
$$42 \times 2.53 \text{ ft} = 106.26 \text{ ft}$$

Pace is good to about 3% $\Rightarrow$ about ± 3 ft.

Report 106 ± 3 ft. The extra digits are decoration, not confidence.

Check your method against something you trust

Four stall lengths taped on the ground, then measured off the orthomosaic. The difference is what you report.



What you report

The worst miss is 2.8 in, so on this lot the orthomosaic is good to about ± 3 in (7 cm).
Not the 1 cm pixel you can see.

Measuring twice is not a check

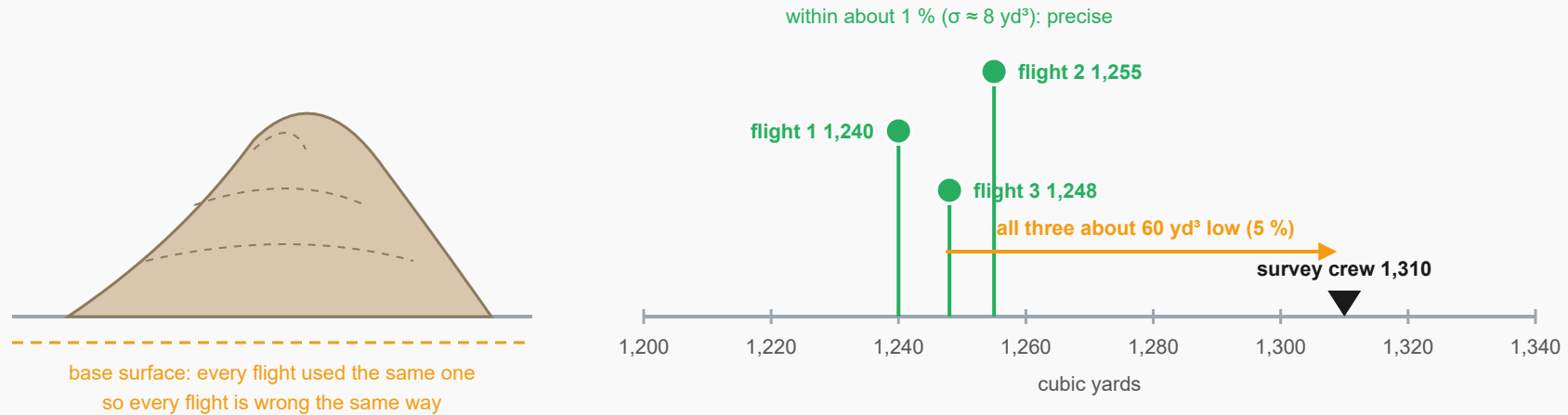
The same tool again only tests precision.
A tape, a survey, or an as-built drawing tests accuracy.

Your method is as good as its worst miss against a source you trust.

Your method is as good as its worst miss against a source you trust.

Three flights agree. That does not make them right.

One stockpile flown three days running, then measured by a survey crew. Volumes in cubic yards.



What repeating the flight showed

That the method is repeatable. Three flights within 1 % of each other say nothing about whether any of them is close to the truth.

What the survey crew showed

A 60 yd^3 bias, about six truckloads, that no number of repeat flights would have found. Only a different source tests accuracy.

Illustrative figures, not a real survey.

Three flights within 1 %, the survey 5 % away. Repeating tested precision; only the survey tested accuracy.

Is that a check?

You measured the lot twice in QGIS and got the same number both times.

What to do Thursday

You measure the same lot three ways. Each one's check is a **different** method.

Method	Its precision comes from	Its check
Pacing	Your stride and the surface	The taped 100 ft
Google Maps	How steadily you trace	The orthomosaic, or a tape
Orthomosaic in QGIS	How carefully you trace	Taped stalls; surveyed points

Pace your calibration distance **three times** and bring your $\bar{x}$ and σ to [Measurements and Methods](#).